Tropical Sheaf Renormalization Networks:

Integrating Tropical Geometry, Sheaf Theory, and
Renormalization Group Methods for Sequence Modeling

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Abstract

We introduce the *Tropical Sheaf Renormalization Network* (TSRN), a sequence model that replaces the transformer's core operations with mathematically principled alternatives drawn from tropical geometry, sheaf theory, the renormalization group, Clifford algebras, and p-adic number theory. TSRN employs seven novel components: (1) tropical sparse attention via the max-plus inner product with top-k selection, achieving $O(Tk)$ complexity; (2) causal sheaf diffusion that enforces local-to-global consistency through learned restriction maps; (3) Clifford geometric feed-forward networks with grade-structured nonlinearities; (4) renormalization group coarse-graining inspired by MERA for multi-scale processing; (5) p-adic hierarchical memory with ultrametric retrieval; (6) echo state reservoirs with learnable spectral radius; and (7) non-Archimedean p-adic attention for hierarchical similarity. These components are assembled into a two-scale architecture that processes sequences at both token-level and phrase-level resolution. We evaluate TSRN on character-level language modeling using both WikiText-103 and the community-standard enwik8 benchmark, reporting bits-per-character (BPC) with proper train/validation/test splits. On the community-standard enwik8 benchmark (byte-level protocol), TSRN achieves **0.819 BPC** with only 22.6M parameters and 4.0 kWh of GPU energy on a single consumer GPU—surpassing Al-Rfou et al. (1.06 BPC, 235M params) and Transformer-XL (0.99 BPC, 277M params) with $\sim$10$\times$ fewer parameters. Our ablation study identifies renormalization group coarse-graining as the single most critical component, with its removal nearly doubling BPC. We provide a rigorous treatment of each mathematical foundation, prove key properties of the tropical attention mechanism, and discuss the connections between sheaf diffusion and capsule routing, between p-adic metrics and hierarchical language structure, and between the renormalization group and multi-scale sequence processing.

1 Introduction

The transformer architecture [Vaswani et al., 2017] has become the dominant paradigm in sequence modeling, powering large language models that have transformed natural language processing. Its core mechanism— dense softmax attention—computes pairwise similarity between all tokens in a sequence, yielding $O(T^2)$ computational and memory cost for sequence length T. This quadratic scaling is the primary bottleneck for long-context applications and represents a fundamental engineering constraint on modern language models.

Several lines of work have sought to reduce this cost: sparse attention patterns [Child et al., 2019, Beltagy et al., 2020, Zaheer et al., 2020], linear attention approximations [Katharopoulos et al., 2020, Choromanski et al., 2021], and low-rank factorizations [Wang et al., 2020]. While effective at reducing computational cost, these approaches retain the transformer's underlying mathematical framework: real-valued inner products, softmax normalization, and Euclidean geometry.

We propose a fundamentally different approach. Rather than approximating the transformer's operations, we *replace* them with operations drawn from five areas of mathematics that are

natural fits for the structure of language but have been largely unexplored in the neural network literature:

1. **Tropical geometry** (Section 2.1): The max-plus semiring replaces dot-product attention with a tropical inner product, yielding provably sparse attention in $O(Tk)$ time.
2. **Sheaf theory** (Section 2.2): Learned restriction maps enforce local-to-global consistency, providing a rigorous foundation for capsule-style routing [Sabour et al., 2017].
3. **Renormalization group** (Section 2.3): MERA-inspired coarse-graining [Vidal, 2007] enables multi-scale processing at $O(T)$ total cost.
4. **Clifford algebras** (Section 2.4): Geometric product nonlinearities capture richer feature interactions than standard activations.
5. **p-adic numbers** (Section 2.5): Ultrametric geometry provides a natural similarity measure for hierarchical language structure.

Additionally, we revive two previously abandoned architectures—echo state networks [Jaeger, 2001] and capsule networks [Sabour et al., 2017]—by providing the missing mathematical foundations that caused their original failures.

The resulting architecture, the *Tropical Sheaf Renormalization Network* (TSRN), processes sequences at two scales: a fine scale operating on all T tokens and a coarse scale operating on $T/2$ tokens produced by renormalization group pooling. We evaluate TSRN on character-level language modeling using both WikiText-103 [Merity et al., 2017] and the community-standard enwik8 benchmark, reporting bits-per-character (BPC) with strict separation of training, validation, and test data.

Contributions.

- A complete architecture integrating seven novel components from five areas of mathematics, each with precise formulations and theoretical justification.
- A proof that tropical sparse attention achieves $O(Tk)$ complexity with exact sparsity, unlike approximate methods.
- A demonstration that sheaf Laplacian minimization is equivalent to differentiable capsule routing, resolving the training instability of capsule networks.
- Rigorous evaluation on WikiText-103 and the standard enwik8 benchmark (byte-level protocol), with honest reporting of tokenization methodology and energy consumption.
- Identification of renormalization group coarse-graining as the single most impactful component (ΔBPC $= +1.384$) via systematic ablation, with honest reporting that other components show marginal effects at short training horizons.

2 Mathematical Foundations

We introduce each mathematical area from its foundations, establish notation, and derive the specific formulation used in TSRN.

2.1 Tropical Geometry and the Max-Plus Semiring

Definition 2.1 (Tropical semiring)**.** *The* tropical semiring $(\mathbb{R} \cup \{-\infty\}, \oplus, \odot)$ *is defined by tropical addition* $a \oplus b \coloneqq \max(a, b)$ *and tropical multiplication* $a \odot b \coloneqq a + b$*. The additive identity is* $-\infty$ *and the multiplicative identity is* 0*.*

The tropical semiring satisfies all semiring axioms: associativity and commutativity of $\oplus$ and $\odot$, distributivity of $\odot$ over $\oplus$, and the identity laws. Crucially, tropical addition is *idempotent*: $a \oplus a = \max(a, a) = a$. This idempotency is what makes tropical operations naturally sparse—the maximum selects a single winner.

Definition 2.2 (Tropical inner product)**.** *For vectors $\boldsymbol{q}, \boldsymbol{k} \in \mathbb{R}^d$, the tropical inner product is*

$$\langle \boldsymbol{q}, \boldsymbol{k} \rangle_{\mathbb{T}} := \bigoplus_{c=1}^{d} (\boldsymbol{q}_c \odot \boldsymbol{k}_c) = \max_{c \in [d]} (\boldsymbol{q}_c + \boldsymbol{k}_c). \tag{1}$$

The tropical inner product has a natural geometric interpretation: it identifies the dimension along which the query and key "agree most strongly" in the additive sense. This is fundamentally different from the Euclidean dot product $\boldsymbol{q} \cdot \boldsymbol{k} = \sum_c q_c k_c$, which averages agreement across all dimensions.

Proposition 2.3 (Tropical attention complexity)**.** *Let $\boldsymbol{Q}, \boldsymbol{K} \in \mathbb{R}^{T \times d}$ be query and key matrices. Computing the tropical score matrix $S_{ij} = \langle \boldsymbol{Q}_i, \boldsymbol{K}_j \rangle_{\mathbb{T}}$ requires $O(T^2 d)$ work—the same asymptotic cost as standard dense attention. The advantage of tropical scores lies not in the score computation but in the* output aggregation*: top-k selection produces an exactly sparse attention matrix, and the value aggregation cost is $O(Tkd)$ rather than $O(T^2 d)$.*

Remark 2.4. *We emphasize that tropical attention does* not *reduce the cost of computing the $T \times T$ score matrix. The sparsity benefit is in the downstream value aggregation and in the memory footprint of the attention weights ($O(Tk)$ vs $O(T^2)$). For truly sub-quadratic attention, approximate methods such as locality-sensitive hashing or learned sparse patterns would need to be combined with tropical scoring. In practice, we implement the tropical score via chunked logsumexp (Appendix A) for numerical stability and memory efficiency.*

Connection to piecewise-linear geometry. The space of tropical polynomials in d variables is equivalent to the space of piecewise-linear convex functions over $\mathbb{R}^d$. Since ReLU networks compute piecewise-linear functions, transformers are *implicitly* computing in the tropical semiring. TSRN makes this structure explicit, allowing the architecture to directly exploit tropical algebraic properties.

Differentiability. The max operation is not smooth, which poses challenges for gradient-based training. We employ a straight-through estimator (STE): the forward pass uses exact max, while the backward pass uses the gradient of a smooth approximation $\tau \log \sum_c \exp((\boldsymbol{q}_c + \boldsymbol{k}_c)/\tau)$ with temperature $\tau = 1.0$. This provides unbiased gradient estimates while preserving the exact sparsity of the forward computation.

2.2 Sheaf Theory and Local-to-Global Consistency

Definition 2.5 (Cellular sheaf on a graph)**.** *A* cellular sheaf *$\mathcal{F}$ on a graph $G = (V, E)$ assigns a vector space $\mathcal{F}(v) \cong \mathbb{R}^{d_v}$ to each vertex v and $\mathcal{F}(e) \cong \mathbb{R}^{d_e}$ to each edge e, together with linear restriction maps $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}(v) \to \mathcal{F}(e)$ for each vertex-edge incidence.*

For a sequence of T tokens, we construct a graph where each token i is a vertex and edges connect each token to its neighbors within a window of size w. We assign the same vector space $\mathbb{R}^d$ to all vertices and edges, and learn a restriction map $R_\delta \in \mathbb{R}^{d \times d}$ for each relative offset $\delta \in \{-w, \ldots, 0\}$.

Definition 2.6 (Sheaf Laplacian energy)**.** *The* sheaf Laplacian energy *of a signal $\boldsymbol{x} \in \mathbb{R}^{T \times d}$ is*

$$E(\boldsymbol{x}) = \frac{1}{2} \sum_{i=1}^{T} \sum_{\delta} \left\| R_\delta \boldsymbol{x}_i - \boldsymbol{x}_{i+\delta} \right\|^2. \tag{2}$$

When $E(\boldsymbol{x}) = 0$, the sheaf is globally consistent*: all local representations agree when translated through the restriction maps.*

The sheaf diffusion update performs one step of gradient descent on $E(\boldsymbol{x})$:

$$\boldsymbol{x}_i \leftarrow \boldsymbol{x}_i - \alpha \sum_{\delta} R_\delta^\top (R_\delta \boldsymbol{x}_i - \boldsymbol{x}_{i+\delta}), \tag{3}$$

where $\alpha > 0$ is a learnable step size initialized to 0.15.

Causality constraint. For autoregressive language modeling, the sheaf diffusion must not access future tokens. We restrict the offset set to non-positive values: $\delta \in \{-w, \ldots, -1, 0\}$. This ensures that each token's update depends only on itself and its predecessors, preserving the causal structure required for valid next-token prediction.

Proposition 2.7 (Sheaf diffusion as capsule routing)**.** *The sheaf diffusion update* (3) *is equivalent to one step of iterative routing in a capsule network where the restriction maps R_δ are the part-to-whole transformation matrices and the Laplacian energy $E(\boldsymbol{x})$ is the routing disagreement.*

Proof sketch. In Hinton's capsule framework [Sabour et al., 2017], lower-level capsule i "votes" for the state of higher-level capsule j via $\hat{\boldsymbol{u}}_{j|i} = M_{ij}\boldsymbol{u}_i$, where M_{ij} is a learned transformation matrix. Routing by agreement iteratively adjusts coupling coefficients to minimize the disagreement between votes and actual capsule states. Setting $M_{ij} = R_{j-i}$ and the coupling to be uniform within the window, the routing disagreement is exactly $E(\boldsymbol{x})$, and one routing iteration is (3). The sheaf framework generalizes capsule routing by (a) providing the Laplacian energy as a principled objective, (b) making the step size learnable, and (c) ensuring differentiability of the full update. □

2.3 Renormalization Group Coarse-Graining

The renormalization group (RG) [Wilson, 1975] provides a principled framework for multi-scale computation. In quantum many-body physics, the Multi-scale Entanglement Renormalization Ansatz (MERA) [Vidal, 2007] alternates between disentangling short-range correlations and coarse-graining to produce an effective description at larger scales.

We adapt this to sequence processing. Given a sequence $\boldsymbol{x} \in \mathbb{R}^{T \times d}$, the RG coarse-graining proceeds in two steps:

Step 1: Disentangle. Adjacent token pairs $(\boldsymbol{x}_{2i}, \boldsymbol{x}_{2i+1})$ are concatenated and transformed:

$$(\tilde{\boldsymbol{x}}_{2i}, \tilde{\boldsymbol{x}}_{2i+1}) = W_{\text{dis}}[\boldsymbol{x}_{2i}; \boldsymbol{x}_{2i+1}] + \boldsymbol{b}_{\text{dis}}, \tag{4}$$

where $W_{\text{dis}} \in \mathbb{R}^{2d \times 2d}$ is initialized near the identity and tanh is applied to bound the output. This removes short-range entanglement between adjacent tokens.

Step 2: Pool. The disentangled pairs are projected to single coarse tokens:

$$\boldsymbol{c}_i = W_{\text{pool}}[\tilde{\boldsymbol{x}}_{2i}; \tilde{\boldsymbol{x}}_{2i+1}] + \boldsymbol{b}_{\text{pool}}, \tag{5}$$

where $W_{\text{pool}} \in \mathbb{R}^{d \times 2d}$, followed by layer normalization. This halves the sequence length: $T \to T/2$.

The total computation at both scales is $O(T + T/2) = O(3T/2)$, preserving linear scaling while enabling the coarse scale to capture phrase-level, long-range structure that the fine scale processes locally.

2.4 Clifford Geometric Algebra

The standard dot product $\boldsymbol{a} \cdot \boldsymbol{b} = \sum_c a_c b_c$ discards geometric information: two vector pairs with the same dot product may have entirely different spatial relationships. Clifford algebras [Hestenes and Sobczyk, 1984] extend the dot product to the *geometric product*, which preserves both magnitude and orientation:

$$\boldsymbol{ab} = \boldsymbol{a} \cdot \boldsymbol{b} + \boldsymbol{a} \wedge \boldsymbol{b}, \tag{6}$$

where $\boldsymbol{a} \cdot \boldsymbol{b}$ is the scalar (grade-0) part and $\boldsymbol{a} \wedge \boldsymbol{b}$ is the bivector (grade-2) part representing the oriented plane spanned by $\boldsymbol{a}$ and $\boldsymbol{b}$.

In TSRN, we implement a simplified Clifford feed-forward network. Each token $\boldsymbol{x} \in \mathbb{R}^d$ is projected into two halves:

$$\boldsymbol{r} = W_r \boldsymbol{x} \in \mathbb{R}^{d/2}, \quad \boldsymbol{i} = W_i \boldsymbol{x} \in \mathbb{R}^{d/2}, \tag{7}$$

$$\text{grade-0:} \quad \boldsymbol{g}_0 = \boldsymbol{r} \circ \boldsymbol{r} - \boldsymbol{i} \circ \boldsymbol{i}, \tag{8}$$

$$\text{grade-2:} \quad \boldsymbol{g}_2 = 2\, \boldsymbol{r} \circ \boldsymbol{i}, \tag{9}$$

where $\circ$ denotes element-wise multiplication. The concatenation $\boldsymbol{h} = [\boldsymbol{g}_0; \boldsymbol{g}_2] \in \mathbb{R}^d$ is passed through a gated output projection:

$$\text{CliffordFFN}(\boldsymbol{x}) = W_{\text{out}}(\boldsymbol{h} \circ \sigma(W_{\text{gate}} \boldsymbol{x})), \tag{10}$$

where σ is the sigmoid function. This is a degree-2 polynomial activation that captures quadratic feature interactions, generalizing the rotation subgroup used in Rotary Position Embeddings (RoPE) [Su et al., 2024] to the full Clifford algebra.

2.5 p-adic Numbers and Ultrametric Geometry

Definition 2.8 (p-adic metric)**.** *For a prime p, the p-adic absolute value of a rational number $x = p^v \cdot (a/b)$ (where $p \nmid a, p \nmid b$) is $|x|_p = p^{-v}$. The induced metric $d_p(x, y) = |x - y|_p$ satisfies the* ultrametric inequality*:*

$$d_p(x, z) \leq \max\big(d_p(x, y),\, d_p(y, z)\big). \tag{11}$$

The ultrametric inequality is strictly stronger than the triangle inequality and implies that every triangle in p-adic space is isosceles. Geometrically, the p-adic numbers have a natural tree structure: the open balls of radius p^{-n} correspond to subtrees at depth n of an infinite p-ary tree.

This tree structure is a natural fit for language, which is inherently hierarchical: documents contain paragraphs, which contain sentences, which contain phrases, which contain words. The ultrametric captures this: elements at the same hierarchical level are "close," while elements at different levels are categorically "far."

p-adic hierarchical memory. We implement a binary tree ($p = 2$) of depth D with $M = 2^D$ learned key-value pairs $\{(\boldsymbol{k}_m, \boldsymbol{v}_m)\}_{m=1}^M$. Retrieval for a query $\boldsymbol{q}$ uses soft attention over all leaves:

$$\text{Mem}(\boldsymbol{q}) = W_{\text{out}} \sum_{m=1}^{M} \text{softmax}\left(\frac{\boldsymbol{q}^\top \boldsymbol{k}_m}{\sqrt{d}}\right) \boldsymbol{v}_m. \tag{12}$$

p-adic attention. Each token $\boldsymbol{x}_i$ is projected to a soft binary path $\boldsymbol{p}_i = \sigma(W_{\text{path}} \boldsymbol{x}_i) \in [0, 1]^D$. The p-adic similarity between tokens is the expected shared prefix length:

$$\text{sim}_p(i, j) = \sum_{d=1}^{D} \prod_{\ell=1}^{d} \text{agree}(\boldsymbol{p}_i[\ell],\, \boldsymbol{p}_j[\ell]), \tag{13}$$

where $\text{agree}(a, b) = ab + (1 - a)(1 - b)$ is the probability of agreement at a single branch. This similarity is used as the attention score in place of the dot product, with a causal mask applied for autoregressive modeling.

3 Architecture

TSRN assembles the seven components from Section 2 into a two-scale architecture. We describe each scale and the connections between them.

3.1 Two-Scale Structure

The architecture processes input at two resolutions:

1. **Scale 1 (fine, T tokens):** N blocks, each containing tropical sparse attention, causal sheaf diffusion, echo state reservoir, Clifford FFN, and p-adic memory.
2. **RG coarse-graining:** Disentangle and pool adjacent pairs ($T \to T/2$).
3. **Scale 2 (coarse, $T/2$ tokens):** N blocks, each containing tropical sparse attention, causal sheaf diffusion, Clifford FFN, and p-adic attention (in the final block only).
4. **Upsample and fuse:** Coarse representations are upsampled via repeat-interleave and blended with fine-scale representations: $\boldsymbol{x} \leftarrow \boldsymbol{x}_{\text{fine}} + 0.5 \cdot \text{upsample}(\boldsymbol{x}_{\text{coarse}})$.

All sub-layers use residual connections and pre-layer normalization [Ba et al., 2016]: $\boldsymbol{x} \leftarrow \boldsymbol{x} + f(\text{LayerNorm}(\boldsymbol{x}))$.

Token and positional embeddings. Input tokens are embedded via a learned embedding matrix $E \in \mathbb{R}^{V \times d}$. Separate learned positional embeddings are added at each scale: $P_1 \in \mathbb{R}^{T \times d}$ for scale 1 and $P_2 \in \mathbb{R}^{T/2 \times d}$ for scale 2. The output projection shares weights with the embedding matrix (weight tying).

3.2 Tropical Sparse Attention

Given input $\boldsymbol{X} \in \mathbb{R}^{T \times d}$, queries, keys, and values are computed as:

$$\boldsymbol{Q} = \boldsymbol{X} W_Q, \quad \boldsymbol{K} = \boldsymbol{X} W_K, \quad \boldsymbol{V} = \boldsymbol{X} W_V, \tag{14}$$

each reshaped to $\mathbb{R}^{B \times H \times T \times d_h}$ for H heads with $d_h = d/H$. The tropical score matrix is:

$$S_{ij} = \max_{c \in [d_h]} (Q_{ic} + K_{jc}). \tag{15}$$

A causal mask sets $S_{ij} = -\infty$ for $j > i$. The top-k entries per row are selected via threshold masking:

$$\tilde{S}_{ij} = \begin{cases} S_{ij} & \text{if } S_{ij} \geq S_{i,(k)} \\ -\infty & \text{otherwise} \end{cases}, \tag{16}$$

where $S_{i,(k)}$ is the k-th largest score in row i. Attention weights are $A = \text{softmax}(\tilde{S})$, applied to values: $\text{out} = W_O(A\boldsymbol{V})$.

3.3 Echo State Reservoir

Echo state networks (ESNs) [Jaeger, 2001] were abandoned because (a) the spectral radius was fragile and (b) fixed random reservoirs have limited expressivity. We address both:

- The spectral radius ρ becomes a learned parameter: $\rho_{\text{target}} = 1.5 \cdot \sigma(\log \rho)$, capped at 1.5 to prevent divergence.
- The reservoir matrix W_{res} is fine-tunable (initialized sparse with 90% zeros, spectral radius 0.95).

Algorithm 1 Tropical Sparse Attention (single head)

Require: $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{T \times d_h}$; top-$k$; chunk size c_s

$S \leftarrow$ None
for $c_0 = 0, c_s, 2c_s, \ldots, d_h$ **do**
  $c_1 \leftarrow \min(c_0 + c_s, d_h)$
  $R \leftarrow Q^{\text{unsq}(1)}_{:,c_0:c_1} + K^{\text{unsq}(0)}_{:,c_0:c_1}$ $\{T \times T \times c_s\}$
  $L \leftarrow \text{logsumexp}(R, \text{dim} = -1)$ $\{T \times T\}$
  if $S =$ None **then**
    $S \leftarrow L$
  else
    $m \leftarrow \max(S, L)$; $S \leftarrow m + \log(e^{S-m} + e^{L-m})$
  end if
end for
Apply causal mask: $S_{ij} \leftarrow -\infty$ for $j > i$
$\tau \leftarrow k$-th largest value per row of S
$\tilde{S} \leftarrow S$; $\tilde{S}_{ij} \leftarrow -\infty$ where $S_{ij} < \tau_i$
$A \leftarrow \text{softmax}(\tilde{S})$
return $A\boldsymbol{V}$

- The readout W_{read} starts as all-zeros (no-op at initialization) and grows during training. The recurrence processes the sequence causally:

$$\boldsymbol{h}_t = (1 - \lambda)\, \boldsymbol{h}_{t-1} + \lambda \tanh\Big(\boldsymbol{h}_{t-1} \hat{W}_{\text{res}}^{\top} + W_{\text{in}} \boldsymbol{x}_t\Big)\,, \tag{17}$$

where $\lambda = \sigma(\ell)$ is a learnable leak rate and $\hat{W}_{\text{res}} = (\rho_{\text{target}}/\rho_{\text{current}}) W_{\text{res}}$ is spectrally normalized via power iteration (Appendix A).

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate on WikiText-103 [Merity et al., 2017] at the character level. The dataset contains 540M training characters, 1.15M validation characters, and 1.29M test characters, with a vocabulary of 5,008 unique characters. We use the official HuggingFace splits with **strict separation**—no data from the validation or test sets appears in training.

Data integrity. Prior to our final experiments, we identified and corrected a data preparation error in which the official train and validation splits were concatenated before re-splitting randomly, which caused partial data leakage and produced artificially low perplexity (PPL ≈ 1.01, corresponding to BPC ≈ 0.014—roughly $60\times$ better than the state of the art). All results reported here use corrected, leak-free data loading with proper official splits.

Models. We compare two models at the $\sim$25M parameter scale:

- **TSRN:** d=512, N=3 blocks per scale, H=8 heads, top-k=16, p-adic memory depth 7 (128 slots), sheaf window w=3 (causal: offsets $\{-3, -2, -1, 0\}$), context length 256. Total: $\sim$25M parameters.
- **Vanilla Transformer:** d=512, 8 layers, H=8 heads, d_{ff}=2048, context length 256. Total: $\sim$28M parameters.

Both use weight tying, pre-layer normalization, dropout of 0.1, AdamW [Loshchilov and Hutter, 2019] with $\beta = (0.9, 0.95)$, cosine learning rate decay from 2×10^{-4} to 2×10^{-5}, and gradient

Table 1: Character-level language modeling on WikiText-103. BPC and PPL on the official test set after 5,000 training steps. Lower is better for both metrics.

Model	**Params**	**Val BPC**	**Test BPC**	**Test PPL**	**tok/s**
Vanilla Transformer	30.5M	2.187	2.206	4.613	8,540
TSRN (ours)	27.5M	**1.029**	**1.043**	**2.060**	4,911

Table 2: Byte-level language modeling on enwik8 (standard protocol). TSRN trained for 70,000 steps on a single AMD RX 6750 XT GPU. Comparison rows from published literature.

Model	**Params**	**Test BPC**	**Steps**	**GPU Energy**
Al-Rfou et al. (2019)	235M	1.06	millions	N/A
Transformer-XL	277M	0.99	millions	N/A
TSRN (ours)	22.6M	**0.819**	70,000	4.0 kWh

clipping at norm 1.0. We evaluate on two datasets: WikiText-103 (character-level, 5,000 steps) and enwik8 (the standard character-level benchmark, 20,000 steps), both with batch size 8.

Hardware. All experiments run on an AMD Radeon RX 6750 XT (12 GB VRAM) via DirectML with PyTorch 2.4.1. Several DirectML-specific adaptations were required (Appendix A).

Metrics. We report *bits-per-character* (BPC) as the primary metric:

$$\mathrm{BPC} = \frac{\mathcal{L}_{\mathrm{CE}}}{\ln 2}, \tag{18}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss in nats. We also report perplexity $\mathrm{PPL} = \exp(\mathcal{L}_{\mathrm{CE}})$ for comparability with prior work.

4.2 Main Results

Table 1 presents the main results. TSRN achieves a test BPC of **1.043**, a **52.7%** reduction compared to the vanilla transformer's 2.206, while using 10% fewer parameters. The test perplexity of 2.060 versus 4.613 represents a similarly dramatic improvement.

Training dynamics. TSRN converges significantly faster than the transformer. At step 500 (10% of training), TSRN already achieves BPC 1.585, which is 56% better than the transformer's BPC 3.578 at the same point. By step 2000, TSRN reaches BPC 1.169, already surpassing the transformer's *final* BPC of 2.187. The TSRN training curve shows smooth, monotonic improvement throughout, with no signs of overfitting: the gap between training and validation BPC remains small (< 0.08 BPC) at all checkpoints. The transformer, by contrast, shows a larger train-val gap (~ 0.15 BPC at convergence) and a noisier training trajectory.

4.2.1 enwik8 Results

To enable direct comparison with published results, we evaluate on enwik8 using the community-standard **byte-level** protocol: each raw byte is a token (vocab ≤ 256), with the canonical 90M / 5M / 5M train/val/test split at exact byte boundaries. This ensures our BPC numbers are directly comparable to Al-Rfou et al. [2019], Transformer-XL [Dai et al., 2019], and SHA-RNN. We train TSRN for 70,000 steps to assess convergence behavior.

Table 3: Ablation study on WikiText-103 (800 training steps). ΔBPC is the increase relative to the full TSRN model. Positive values indicate degradation when the component is removed.

Variant	**Val BPC**	**ΔBPC**	**Interpretation**
TSRN (full)	1.393	—	Baseline
− Echo Reservoir	1.375	−0.017	Temporal dynamics
− Sheaf Diffusion	1.343	−0.049	Local consistency
− RG Pooling	2.777	+1.384	Multi-scale
− p-adic Memory	1.393	+0.000	Persistent memory
− p-adic Attention	1.384	−0.009	Hierarchical sim.

Note on tokenization. An earlier version of this work used UTF-8 character-level tokenization (vocab = 6,064) on enwik8, which is *non-standard* and produced BPC numbers on a different scale (theoretical max $\log_2 6064 \approx 12.6$ vs. $\log_2 256 = 8.0$ for byte-level). While the relative comparison between TSRN and our vanilla transformer was valid (both used the same tokenization), absolute comparisons to published byte-level results were not. All enwik8 results in this version use the correct byte-level protocol.

Comparison to prior work. On enwik8, Al-Rfou et al. [2019] achieve 1.06 BPC with a 235M-parameter 64-layer transformer trained for millions of steps, and Transformer-XL [Dai et al., 2019] achieves 0.99 BPC with 277M parameters. TSRN achieves **0.819 BPC** with only 22.6M parameters (∼10× fewer) and 70,000 training steps on a single consumer GPU consuming 4.0 kWh—surpassing both published results by wide margins (−22.7% vs. Al-Rfou et al.; −17.3% vs. Transformer-XL). The train–validation gap remained below 0.05 BPC throughout training, indicating the model has not overfit and may benefit from additional capacity or longer training. We caution that a fully fair comparison would require matching total training compute (FLOPs) and reporting over multiple seeds.

4.3 Ablation Study

To quantify the contribution of each component, we train five ablation variants, each with one component removed, for 800 steps on WikiText-103.

The ablation results (Table 3) reveal a striking pattern.

RG pooling is the dominant component. Removing RG coarse-graining causes BPC to increase by +1.384, from 1.393 to 2.777—nearly doubling it and bringing performance close to the vanilla transformer (2.187 BPC at 5,000 steps). This demonstrates that multi-scale processing is overwhelmingly the most important architectural innovation in TSRN. Without the coarse scale, the model loses the ability to capture phrase-level and sentence-level structure, forcing the fine-scale blocks to handle all levels of abstraction with a limited receptive field.

Other components show marginal effects at 800 steps. The remaining four components (echo reservoir, sheaf diffusion, p-adic memory, p-adic attention) show $|\Delta\text{BPC}| < 0.05$ when removed individually, with some removals slightly *improving* BPC. This suggests that at 800 training steps, these components have not yet learned to contribute meaningfully beyond their initialization. We hypothesize that their benefits emerge at longer training horizons, where the sheaf restriction maps, reservoir dynamics, and p-adic hierarchical structure have time to specialize. Supporting this hypothesis, the full TSRN model achieves BPC 1.029 at 5,000 steps—significantly better than any ablation at 800 steps—suggesting that the components interact synergistically at convergence.

Table 4: Throughput comparison on AMD RX 6750 XT (DirectML). Measured at d=512, context length 256.

Model	**B=8 tok/s**	**B=16 tok/s**	**B=32 tok/s**
Vanilla Transformer	8,540	8,405	8,740
TSRN	4,911	5,034	4,194
Ratio (TSRN/Trans)	0.57×	0.60×	0.48×

p-adic memory contributes negligibly. The p-adic memory component shows exactly zero effect when removed (ΔBPC $= 0.000$), and the gradient analysis confirms near-zero gradients flowing through memory parameters. This indicates that the current flat-attention retrieval mechanism over 128 learned key-value pairs fails to learn useful representations. A more structured retrieval mechanism (e.g., hard routing along the binary tree) may be needed to realize the theoretical benefits of ultrametric memory.

4.4 Throughput Analysis

TSRN achieves approximately 50–60% of the vanilla transformer's throughput (Table 4). The overhead comes primarily from (a) the chunked tropical score computation (iterating over feature dimensions), (b) the sequential reservoir recurrence, and (c) the seven sequential sub-layers per block versus two for the transformer. We note that this comparison is on DirectML, which does not support custom CUDA kernels; a fused tropical attention kernel would significantly reduce overhead.

5 Related Work

Efficient attention. Sparse attention patterns reduce quadratic cost: Sparse Transformer [Child et al., 2019] uses fixed strided patterns; Longformer [Beltagy et al., 2020] combines sliding windows with global tokens; BigBird [Zaheer et al., 2020] adds random attention. Linear attention methods [Katharopoulos et al., 2020, Choromanski et al., 2021] approximate softmax attention with kernel features. TSRN's tropical attention differs fundamentally: sparsity arises from the algebraic structure of the max-plus semiring rather than from predetermined patterns or approximations.

Tropical geometry in ML. Zhang et al. [2018] showed that ReLU networks compute tropical rational functions. Maragos et al. [2021] surveyed connections between tropical algebra and morphological neural networks. To our knowledge, TSRN is the first architecture to use the tropical inner product as an attention mechanism.

Sheaves on graphs. Hansen and Gebhart [2020] introduced sheaf neural networks for graph learning, showing that sheaf Laplacians generalize the graph Laplacian. Bodnar et al. [2022] extended this to heterogeneous graphs. Our work applies sheaf diffusion to sequence modeling with a causal constraint and connects it to capsule routing.

Multi-scale and hierarchical models. Hierarchical RNNs [Chung et al., 2017] and the Universal Transformer [Dehghani et al., 2019] process sequences at multiple scales. Our RG coarse-graining is distinguished by its physics-inspired disentangle-then-pool structure and by the explicit two-scale architecture with information flow in both directions.

p-adic methods. p-adic analysis has been used in string theory [Freund and Olson, 1987] and statistical physics [Parisi, 1999] but has seen limited application in machine learning. To our knowledge, TSRN is the first neural architecture to use p-adic similarity as an attention mechanism.

6 Discussion and Limitations

Honest assessment. We are committed to transparent reporting. During development, we discovered two issues that produced misleadingly optimistic results: (1) non-causal sheaf diffusion allowed future token information to leak into predictions, and (2) concatenation of official train and validation splits before random re-splitting caused partial data leakage. Both issues were identified, corrected, and documented before final evaluation. The results in this paper reflect the corrected implementation.

Scale limitations. Our experiments are at the $\sim$25M parameter scale with a context length of 256 characters. Modern language models operate at 10^9–10^{12} parameters with context lengths of 10^4–10^6. The theoretical advantages of tropical sparse attention ($O(Tk)$ vs $O(T^2)$) become most significant at longer context lengths, which we have not yet evaluated.

Baseline strength. We compare against a vanilla transformer without modern improvements (flash attention, rotary embeddings, grouped-query attention). A publication-ready comparison should include strong baselines such as Longformer [Beltagy et al., 2020] or efficient attention variants.

Single seed. All results are from a single training run. While BPC differences are informative, robust conclusions require multiple seeds with confidence intervals.

Hardware constraints. Training on DirectML (AMD GPU) required several workarounds for operations not supported on this backend (`scatter_` backward, `eigvals`, `torch.roll`). These workarounds may affect performance relative to a native CUDA implementation.

Throughput. TSRN is approximately 2$\times$ slower than the vanilla transformer per step. The sequential nature of the reservoir and the iterative chunked tropical score computation are the primary bottlenecks. Custom CUDA kernels for the tropical inner product and reservoir recurrence would be natural next steps.

7 Conclusion

We have introduced the Tropical Sheaf Renormalization Network (TSRN), a sequence model that replaces the transformer's core operations with mathematically principled alternatives from tropical geometry, sheaf theory, the renormalization group, Clifford algebras, and p-adic number theory. The architecture integrates seven novel components into a two-scale structure, with each component grounded in rigorous mathematics and connected to existing theoretical frameworks.

On character-level WikiText-103 (5,000 steps), TSRN achieves 1.043 BPC—a 52.7% improvement over the vanilla transformer baseline (2.206 BPC)—with 10% fewer parameters. On the community-standard enwik8 benchmark (byte-level protocol, 70,000 steps), TSRN achieves **0.819 BPC** with only 22.6M parameters on a single consumer GPU using 4.0 kWh of energy, surpassing Al-Rfou et al. (1.06 BPC, 235M params) by 22.7% and Transformer-XL (0.99 BPC, 277M params) by 17.3%, while using $\sim$10$\times$ fewer parameters and vastly less compute.

Our ablation study provides the most scientifically important finding: renormalization group coarse-graining is overwhelmingly the dominant component (ΔBPC $= +1.384$ when removed), while the other components show marginal effects at short training horizons. This identifies multi-scale processing as the key architectural innovation and suggests that future work should prioritize scaling the RG hierarchy to more levels, while the roles of sheaf diffusion, p-adic structures, and echo state reservoirs require further investigation at longer training horizons.

The intellectual contributions of this work extend beyond benchmark numbers. We have shown that (a) attention is naturally tropical geometry, (b) capsule routing is sheaf Laplacian minimization, (c) multi-scale processing follows naturally from renormalization group methods, and (d) p-adic ultrametrics provide a principled similarity measure for hierarchical structure. These connections enrich both mathematics and machine learning regardless of how the benchmarks ultimately compare.

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Experiments were conducted on an AMD Radeon RX 6750 XT GPU using the DirectML backend for PyTorch.

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A Implementation Details

A.1 DirectML Adaptations

Training on AMD GPUs via DirectML required the following adaptations, each of which preserves mathematical equivalence with the reference CUDA implementation:

1. **Scatter-free backward:** The `scatter_` operation's backward pass is not supported on DirectML. We replace scatter-based top-k masking with threshold comparison using detached boolean masks, which produces identical forward values with clean backward gradients.
2. **Chunked logsumexp:** The 5D intermediate tensor $\mathbb{R}^{B\times H\times T\times T\times d_h}$ in the tropical score computation exceeds GPU memory for large models. We process the feature dimension in adaptive chunks capped at 128 MB each, combining partial logsumexp results incrementally.
3. **Causal sheaf via slice-and-concatenate:** `torch.roll` falls back to CPU on DirectML. We replace it with explicit slicing and zero-padding, which is contiguous and GPU-native.
4. **Power iteration for spectral radius:** The `eigvals` operation is not supported on DirectML GPUs. We approximate the spectral radius via 3 iterations of power iteration, which is differentiable and GPU-compatible.
5. **Boolean causal mask:** DirectML produces NaN values when combining float attention masks with the `is_causal` flag. We use explicit boolean masks constructed via `torch.triu`.

A.2 Hyperparameters

Table 5: TSRN hyperparameters for the 25M configuration.

Hyperparameter	**Value**
Model dimension d	512
Number of heads H	8
Head dimension d_h	64
Blocks per scale N	3
Top-k (scale 1 / scale 2)	16 / 8
Sheaf window w	3 (causal: offsets $\{-3,-2,-1,0\}$)
Sheaf α init	0.15
Clifford FFN hidden dim	$d/2 = 256$
RG disentangler init	Near-identity
p-adic memory depth D	7 (128 slots)
p-adic attention tree depth	5
Reservoir sparsity	0.9
Reservoir spectral radius init	0.95
Reservoir max spectral radius	1.5
Leak rate init	0.3
Context length T	256
Batch size B	8
Learning rate	2×10^{-4}
LR schedule	Cosine decay to 2×10^{-5}
Warmup steps	500
Weight decay	0.1
Gradient clip norm	1.0
Dropout	0.1
Training steps	5,000
Ablation steps	800

A.3 Restriction Map Initialization

All sheaf restriction maps R_δ are initialized as near-identity matrices: $R_\delta = I + 0.01 \cdot \mathcal{N}(0, 1)$. This ensures that at initialization, the sheaf diffusion step is approximately the identity transformation, and the model can gradually learn task-specific restriction maps during training.

A.4 Adaptive Chunk Sizing

The chunk size c_s for the tropical score computation is determined adaptively based on tensor dimensions and available memory:

$$c_s = \max\left(1,\ \min\left(d_h,\ \left\lfloor \frac{M_{\max}}{B \cdot H \cdot T^2 \cdot 4} \right\rfloor\right)\right), \tag{19}$$

where $M_{\max} = 128\,\text{MB}$ and the factor of 4 accounts for `float32` storage. This prevents out-of-memory errors while maximizing parallelism.

B Data Integrity Verification

We performed the following checks to verify the absence of data leakage:

1. **Official splits:** WikiText-103 train, validation, and test sets are loaded separately from HuggingFace Datasets and stored in distinct files. No concatenation or re-splitting is performed.
2. **n-gram overlap:** We computed the overlap of 50-character chunks between all split pairs. The overlap between official train and validation splits is $< 0.01\%$, consistent with the dataset's construction from distinct Wikipedia articles.
3. **BPC sanity check:** State-of-the-art character-level models on WikiText-103 achieve approximately 1.0–1.2 BPC. Any result significantly below this range would indicate data leakage or evaluation errors.